Science in the Dock: How Judges Navigate Conflicting Expert Testimonies

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Abstract: The phrase "Science in the Dock" refers to a situation in which scientific discoveries or methods are put on trial, particularly in criminal trials. This occurs when there is an examination of forensic proof, an expert scientific witness, or any particular technique that a scientist used is questioned.

The legal domain now has to deal with the competition of law and science. It tackles the nuanced realm where, as in both civil and criminal disputes, contested claims are negotiated into competing evidences or truths, and judgment is passed not by trained practitioners of scientific approaches. From forensic examination to litigation over climate change, the reliability of the so-called expert's claims can tip the scale toward justice or injustice, imprisonment or freedom, accountability or unaccountability for corporations.

The box marked 'court' is not optimal for a systematic conversation. In like manner to legal research, unlike other science disciplines that thrive on perpetual hypotheses, endless revisions, and ambiguous conclusions, 'science' in law lacks peer-reviewed scrutiny, only having check-listed 'proceedings' that demand either one or the other conclusion: guilty or innocent, liable or not to be considered innocent.

This paper explores into the mechanisms courts use to evaluate expert opinions, from the *Daubert* standard's criteria for reliability to the risks of "junk science" swaying juries. This also explores the intersection of science and law, focusing on how judges navigate conflicting expert testimonies, the limitations of forensic science, and potential reforms to strengthen the judicial evaluation of scientific evidence.

Keywords: Science in the Dock, Expert's testimonies, corroboration of opinion, Navigation of conflict, Daubert standard.

Introduction

The underpinning of the Indian judicial system rests on the maxim *Ei incumbit probatio qui dicit, non qui negat* the burden of proof lies on the one who asserts, not on the one who refutes. This replicates the assumption of goodness until

proven guilty, wherein an individual is considered innocent. Forensic science considered as an essential part of criminal justice.[1]

Admissibility of Scientific Evidence in Criminal Trial

Admissible evidence refers to info presented in court that meets legal standards, meaning it cannot be dismissed based on trivial arguments. Such evidence must hold probative price that is, it must be appropriate of proving or disproving a fact in issue.

To introduce scientific evidence in court, experts employ various test center and fact-finding techniques. An expert witness, usually somebody with specialized knowledge or experience, is called upon to explain and defend the methodology and detections behind the scientific evidence being offered. This expert's testimony can show a momentous role in determining the outcome of a trial, particularly when it worries questions of guilt or innocence. Forensic experts, in particular, often testify about their measures and conclusions during criminal proceedings.

In the dominion of forensic science, the judge's role is vital. The focus should not solely be on whether a new scientific theory is broadly accepted, but on whether the specific application of that theory in the case at hand meets the core evidentiary requirements: authenticity, relevance, and reliability[2].

Experts Testimonies Under the Indian Legal System

Generally, when a person is summoned to court for giving testimony as a witness, he is expected to state only facts and not expected to state only facts and not give opinion. It is the job of the court to verify the all evidence and form an opinion in the case. But there is an exception, the experts are considered as witnesses although they are not actually related to the case. The new evidence act is called Bharatiya Sakshya Adhiniyam (BSA), 2023 which replaced Section 45 and retains the provision for expert opinion under Section 39. When the court needs an opinion in a subject which requires special knowledge, the court calls the

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[1] 'Role of forensic science in crime investigation' (Lloyd Institute of Forensic Science, 6 October, 2025) https://lifs.co.in/blog/role-of-forensic-in-crime-investigation.html#:~:text=The%20role%20of%20forensic%20science,potential%20interconnections%20to%20the%20accused. accessed 17 May 2025

[2] Khyati Rao, Rajiv Ratan Singh, Pradeep Kumar Yadav and Sachin Kumar Tripathi, 'Role of scientific evidence in the judiciary system: A Systematic Review' (2023) 17 (4) IJFMT https://medicopublication.com/index.php/ijfmt/article/view/19945#:~:text=Result%2D%20Forensic%20evidence%20plays%20a,transparency%20and%20accountability%20in%20the accessed 16 May 2025

person who is skilled with special knowledge and experiences in a particular area of subject because without their opinion the court cannot form a correct judgement. The opinion given by the person who is specialized with special knowledge which are not the jurisdiction of layman, is called an expert and the opinion given by them is known as expert opinion.

Example: The court is not sure about the fingerprint of the person 'Y'. That time the court calls a fingerprint expert to find out the same. This person will be known as an expert and the opinion given by that person is relevant and known as expert opinion.

Shortly Discussed the Provisions of Expert Opinion Mentioned Under Indian Evidence Act, 1872

The longstanding Indian Evidence Act, 1872 described the expert opinion under the Section 45 to 51[3]. According to Section 45 a person will be considered as an expert when they have developed their time in learning a special branch of expertise and thus are specially skilled in the subject, which must include special knowledge, observation and practical experiences.

Experts is not judge and also their opinion are not obligatory on the court. Because of an expert opinion, the court does not become functus officio (having performed one's duty'). At the end, the final opinion has to be given by the court. The courts need an expert to form an opinion:

1. Foreign law: When a case involves the interpretation of foreign law, the court may rely on the opinion of an expert in that legal system. The expert must be someone familiar with the laws of the foreign country. Because the court itself cannot take judicial notice of foreign law and depends on expert testimony.
2. Science and Art: The word 'science and Art' are to be broadly constructed. The term 'science' and 'is not limited to fine arts, but having its original senses of handicraft, trade, profession and skill in work.
3. Identity of finger impression: Fingerprint evidence is well thought-out highly consistent in criminal cases for establishing identity.
4. Electronic evidence: When information is conveyed or stored in a computer system and the court requires expert aid to understand or assess it, the matter may be referred to an Examiner of Electronic Evidence, who is well-thought-out an expert of such cases.

[3] Leeza Lowanshi, 'Opinion of Third Person When Relevant' (Jus Corpus, 17 October, 2024) https://www.juscorpus.com/opinion-of-third-person-when-relevant/ accessed 18 May 2025

Other expert evidence areas include fingerprints, brain mapping, polygraphs, DNA, ballistics and many others.

Facts bearing upon opinion of experts: Section 46 of the Indian Evidence Act, 1872 provides that when an expert gives an opinion on a relevant matter, any facts that either support or contradict that opinion also become relevant in court proceedings.

Illustration: If it is necessary to determine whether A has been poisoned, then the fact that A shows symptoms which experts confirm or deny as being consistent with that specific poison becomes relevant to the case.

Value of Expert Evidence Under Other Laws

Scientific evidence is admissible in India.

A. *The Code of Criminal Procedure, 1973*

Most important acknowledged aspect of Forensic Science is DNA and it's testing. There is no official legislation passed w.r.t. to DNA testing in India , but Sec 53 and 54 of Cr.P.C deals with examination of alleged person by a certified medical practitioner on reasonable grounds of inquest. The Indian government and the Law Commission also became aware of the issue, and the Indian Parliamentary Affairs Board has established an advisory committee to provide a complete report on all aspects of DNA testing.

Section 53A authorizes investigating officers to collect DNA samples from individuals accused of rape, with the assistance of registered medical practitioners.

B. *The Identification of Prisoners Act, 1920*

This act allows investigating officers to collect fingerprint and footprint impressions from suspects.

Section 5 specifically allows a magistrate to order a person to be measured or photographed for investigation purposes. If satisfied that it is necessary for an investigation or legal proceeding under the CrPC, the magistrate can issue such an order, requiring the individual to appear at a specified time and place for measurements or photographs to be taken by a police officer.

However, only a First-Class Magistrate is authorized to issue orders for photography.

The Information Technology Act, 2000

This act introduced the concept of electronic evidence into Indian law.

Through Section 92 of the IT Act, the Indian Evidence Act was amended to include "electronic records" within the definition of evidence, thus making digital evidence legally admissible.

While the IT Act does not explicitly define or address "expert opinion" as the Indian Evidence Act does, it implies the need for expert involvement in areas requiring technical expertise.

Section 79A of the Information Technology Act says that The Central Government may notify any department, body or agency of the government as an Examiner of Electronic Evidence. Their role is to provide expert opinion on electronic evidence in any proceeding before a court or authority. This provision gives legal recognition to electronic forensics experts[4].

Judges Navigate the Expert Testimony

A. Appraising the Credentials of the Expert

The Evidence Act of the Expert System 45 Judiciary relates to laws within relationships where an expert's testimony is necessary in areas that contain specialized knowledge, such as science, handwriting, foreign law, or medicine. There is no testimony without an examination quite like the one in focus here, that of an expert's qualifications. In simple terms, pre-exam requirements, curriculum vitae, job history, publications, and the most important in this context, ancillary.

If they are conflicting, the locus of authority, or the court in our case relies on the one expert who has the most appropriate education, specialization, or even years working concerning the subject they argue on. To illustrate, the trained and experienced forensic scientist will overpower the general practitioner on the stand when the case's evidence is too much forward forensic.

B. Determining the Method Employed and Acceptance

After all the necessary qualifications have been approved the next in line is the methodology. In this case the opinion regurgitated should be that of one whose methods is known to be credible and verifiable scientifically, not rely on made up methods. Opinions based on conjectures or not prevalent practices in a particular field – the court will not accept the opinion.

In the case *State of H.P. v. Jai Lal,*[5] the supreme court ruled in this matter that the expert opinions must be drawn from methods proven scientifically. Mixing methods or on approaches, that draws no boundaries, results in a lack of appropriate or trustworthy evidence.

C. Coherence and Verification with Other Facts

[4] Dr. Kusum Chauhan, 'Admissibility and Evidentiary Value of Scientific Evidence: Legislative and Judicial Approach in India' (2023) 8 (1) IJRTI https://ijrti.org/papers/IJRTI2301024.pdf accessed 18 May 2025

[5] (1999) 7 SCC 280

Expert testimony, in isolation, lacks meaningful evidential value. This type of testimony must also be supported by other facts and pieces of evidence related to the case at hand.An expert's conclusion that contradicts proven facts or stands unsupported by any corroborative material may be rejected.

In the case *Murari Lal v. State of M.P,*[6] the court ruled that expert evidence is merely advisory and not conclusive. It must be evaluated along with other evidence.

In the case *Ramesh Chandra Agrawal v. Regency Hospital Ltd.,*[7] The case law suggests that the court is under no obligation to accept an expert opinion which is contrary to established facts or evidence.

Thus, the expert's input is important, but the judgment is based on the overall evidence presented.

D. Cross Examination (Section 146)

As stated in Section 146, the party in opposition also has the right to cross-examine the expert. This is the most important moment in which the credibility of the expert can be verified.

This procedure guarantees that testimony provided by an expert will not be taken as truth blindly and will assist the court in verifying that the expert has acted professionally and rationally.

E. Expert by the Court on Its Own Initiative (Section 165)

In very sophisticated or advanced technical matters, the court might find it hard to deal with conflicting expert opinions. In such cases, the judge is allowed under Section 165 to appoint a specific individual whose expertise is not limited to the sides involved in the dispute to aid the court.

This independent expert does not belong to either side and must give an impartial and credible opinion based on factual data. The judge may also interrogate this expert directly to clarify technical points.

Constitutionality And Admissibility of Expert Testimony

Expert evidence is admissible in court only when the expert is examined as a witness. For such evidence to be considered valid, the expert must clearly explain the basis of their opinion and be available for cross-examination by the opposing party. Without this, the opinion is not admissible.

Legal provisions for the examination of experts are laid out in various statutes. Under Section 293 of the Criminal Procedure Code reports submitted by government scientific experts are admissible as evidence in inquiries, trials, or

[6] (1980) 1 SCC 704

[7] (2009) 9 SCC 709

other judicial proceedings. If the expert is unable to appear in person, the court may allow a qualified and knowledgeable officer from the expert's team, who is familiar with the analysis and findings, to appear on their behalf.

Problems With Their Reforms and Solutions

Problems

A. The judge's inability to apply scientific training to cases

The overwhelming of judges are trained lawyers and have no technological education. Consequently, they lack the requisite skills to properly evaluate the scientific data or the highly sophisticated methodologies which are entwined with the scientific components of trials. Understanding science to this level does not exist for many judges and therefore they are forced to make decisions based on clever cross-examinations and pathetic diatribes which do not demarcate the lies from the reality in the information given by the so-called experts.

B. Determining whose judgment is credible

The court is frequently left to determine which argument coming from expert testimony is the most credible when several experts give contradicting testimonies. Analyzing pieces of evidence in these testimonies is both challenging and unwieldy because of how technical the arguments are. Judges and jurors without deep knowledge of the science underlying the testimony may be faced with superficial indicators like assertiveness, educational qualifications, and even mannerisms instead of actual sound arguments.

C. Subjectivity and prejudice

Despite the impartial stance expert witnesses are expected to adopt, personal biases, financial gain, or affiliations can easily alter the quality of presented evidence. The objectivity achieved by experts may be severely distorted under the undue influence of one party hiring an expert known to actively spin the outcome into a certain interpretation.

D. Different Levels of Competency

There is no single criterion by which all experts can be evaluated. Courts may receive opinions from a spouse with varying competence and, without proper resources, they might fail to identify the difference between a subject-area resource and an inadequate dabbling non-expert. This imbalance leads to the testimony being assigned different values which is not proportional to the weight that should be given.

E. Complicated Nature of the Evidence

Evidence oftentimes incorporates expert analysis which can be based on highly complex scientific methodologies, sophisticated statistical computations, or cutting-edge jargon. This expert capsule may be too much for a member of jury

to comprehend, or even a judge for that matter, which will result in the trimming down of the oversimplified version of sophisticated problems. The critical subtleties which are fundamental to the correct resolution of the case are omitted along with the translation.

F. Boundaries of Scope of Expert Testimony

Disputes and controversies within the law resulting from a definite legal boundary which is going to be set for the range of expert evidence can at times also occur. Every now and then, interdisciplinary experts tend to overreach their boundaries and provide an irrelevant opinion concerning the legal question and issue which is ideally supposed to be handled by the court. It is equally as important for the judge to take notice of, and make use of, the expert evidence presented while ensuring that he or she does not become deprived of the use of judicial sovereignty.

G. Issues with Cross-Examinations

The highly IT related reasoning given by the expert may prove to be impossible to cross-examine for most attorneys. Having a lack of broad understanding may inhibit the attorney's ability to maximize exploitation of the expert's analysis. Such a situation may arise that expert's statement which is actually incorrect goes unchallenged throughout legal scrutiny.

H. Lack of Adaptability

Rapid progress observed in science and technology does not fully integrate into the legal framework. Evidence by the expert may still be considered acceptable or reject without relying on more up-to-date information evidence standards. For example, some forensic techniques that were previously accepted as accurate measures may be discredited later as they had become obsolete, but the judiciary is unable to move past the change in acceptability.

I. Increase of Work Due to Disagreement

In the event that there are highly, strongly differing views shared by different experts, it forces courts to bring in other experts or other hearings which consume time and litigation funding. This creates a scenario where justice becomes inaccessible and financial obligations become amplified for the parties within the dispute.

Reforms & Solutions

A. Strengthening Credibility Assessment of Expert Witnesses

1."Daubert Standard" Adoption: Currently, Indian courts are guided by Section 45 of the Indian Evidence Act (IEA), 1872 which does not provide adequate guidance on expert evidence evaluation. It is suggested that a structured admissibility test such as the Daubert standard (from the USA) which offers

criteria to measure the scientific validity of an expert's evidence be considered. The court considers the following standards:

If a theory or method is capable of being tested and has been tested, has it also been evaluated through peer review. Such as: What are the established or possible margins of error associated with it. And also the existence and maintenance of various standards. Implementing this logic in India, be it through judicial activism or statutory reform, will enable us to administer proper control in the acceptance of expert testimony.

2. Advisors from the Scientific Field or Specialized Science Courts: The diverse technical and scientific evidence may surpass the training judges obtain. Some jurisdictions (France and Germany) have adopted science courts, or have technical advisors bridge this cap. This could be useful in India by:

Setting up specialized science benches in High Courts, or tribunal units for forensics, medicine, and cyber security.

3. Conflict of Interests that Must Be Disclosed: In order to maintain biased credibility-Affiliation, financial interest, or other dealings with the party must be disclosed by experts, and such disclosure must be mandatory.

In relation to the above, courts can consider disallowing or removing the designation of an expert if his/her's bio is marked with a significant conflict of interest; it can be done through section 146 (cross examination) of IE Act, and bias can be brought out in the open which is the motive in question.

B. Objectivity and Bias Reduction

1. Experts Appointed by the Court who Remain Neutral: A judge is recommended to exercise powers within Indian Evidence Act, which empowers the judge to ask questions to uncover the truth under Section 165 of the and the Civil Procedure Code pertains to appointing commissions for local investigations under Order 26, Rule 9.

And appoint such experts as they deem fit to be independent. This type of selection neutralizes partisanship resulting from adversarial engagement or the use of partisan professionals as well as objectivity as it was supported in *State of H.P. v. Jai Lal*[8].

2. More Strict Ethical Standards for Experts: There is no comprehensive code of expert witnesses conduct in place for India. It would be the responsibility of the Bar Council of India or the Law Commission to formulate.

Ethical norms that are set should also cover being unbiased and disclosure admitting non-compliance can be non-evidentiary and falsifying evidence violations indeed can be endorsing accepting section 193 IPC (perjury). There is

[8] (1999) 7 SCC 280

judicial Marked UK Civil Procedure rules Part 35. Such rules require adorning court assistance for party supporting rather than experts.

C. Ensuring Uniform Expertise Standards

1. Accreditation of Experts: Only certified individuals, like those from Forensic Science Laboratories, should be allowed to give expert testimony. Courts are to undertake a credential check to ensure that the evidence given is within the crediential scope of trained and experienced individuals rather than unqualified and pseudoscience dabblers.

2. Pre-Trial Voir Dire (examination of the expert witnesses preliminarily): Borrowed from common law, courts should also undertake voir dire to assess- Qualifications, Methodologies and Relevance of the proposed expert testimony. This avoids unqualified or unreliable testimony from being led at later stages in the proceedings.

D. The Strategy of Dealing with Complex Evidence for the Courts and the Juries

1. Use of Court Experts (Amicus Curiae in Technical Cases): Judges are permitted to appoint the technical expert as amicus curiae in difficult cases, as adviced in *Vikramjit Singh v. State of M.P*[9]. Further, Judicial academies ought to provide literacy training on forensics, digital evidence, including AI.

2. Bringing Clarity into Comprehension: Experts should provide- A technical report, and a summary of such report in a simple language.

This two-fold approach will help the judges and even lawyers analyze and appreciate the evidence.

E. Scope Clarification for Expert Opinion Details

Apply "Relevance" & "Helpful" Criteria Literally: Expert opinion should be heard and accepted only when strictly necessary (per Section 45, IEA). Speculative opinions should be rejected as held in Ramesh Chandra Agrawal v. Regency Hospital.

F. Improving Efficiency of Expert Cross Examination

1. Disclosure of Expert Reports Before Trial: Foster exchange of reports before trial (under Order 18, CPC) so that lawyers prepare effective cross-examinations.

2. Training Lawyers on Technical Cross Examination: Law schools, including NALSAR, ILI, and Bar Associations, should develop courses on advanced forensic science and technical evidence, including mock trials where advanced expert cross-examinations are emphasized.

G. Keeping up with developments

[9] (1991) 4 SCC 62

1. Advanced Precedent in Legal Frameworks: New AI, neuroscience, and digital forensics warrant agile approaches within the law.
2. New Dynamic Precedent: Courts need to adopt international jurisprudence practices (consideration of DNA profiling standards in *Selvi v State of Karnataka*[10]) to strengthen and modernize India's evidential framework.

H. Preventing Delays Due to Conflicting Opinions

To avoid a "battle of experts," courts should limit the number of expert testimonies unless necessary.

Early Case Management: Judges should identify technical disputes early and appoint neutral experts in the pre-trial phase, ensuring focused and timely resolution of scientific issues.

Empirical Insights with Clear Guidance

Ramesh Chandra Agrawal v. Regency Hospital Ltd & Ors[11]: In this landmark case, the court comprehensively examined the role and definition of an expert. It held that and expert must be devotes their time and study to a specific branch of knowledge. Such expertise may be acquired through practice, observation, or careful study. Importantly, Indian legal system does not power the experts to work as Judge or jury.

An expert is not considered a witness of fact like other witnesses; rather, their testimony is advisory in nature. Their responsibility is to provide the judge with the scientific or technical criteria necessary to assess the validity of conclusions presented. Judge independently evaluate the evidence using those criteria. The court also emphasized that no expert can assert with absolute certainty that their opinion is infallible.

Malay Kumar Ganguly v. Dr. Sukumar Mukherjee[12]: In this case, the court held that it is not obligated to accept expert evidence as conclusive, since such evidence is largely advisory in nature. The court retains full authority to independently assess the expert opinions presented by both parties. It must do so with caution, considering relevant legal principles and precedents related to the subject matter of the expert's testimony.

Bant Singh v. Niranjan Singh[13]: In this case, the dispute arose over a property suit in which the plaintiff claimed that the defendant was his sister. The opinion of his brother-in-law who was 80 years old and has good knowledge of his family members was held as the relevant evidence.

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[10] (2010) 7 SCC 263
[11] (2009) 9 SCC 709
[12] (2009) 9 SCC 221
[13] AIR 2008 SC 7162

A. *Supreme Court should issue binding precedents on when expert evidence is conclusive v. corroborative:*

State of Kerala v. Mahesh [14]: Expert opinion alone cannot convict without supporting evidence.

Gade Lakshmi Mangaraju v/s. State Of Andhra Pradesh[15]: This case clarified that the absence of fingerprint evidence does not definitively prove that a person was not present at the scene.

State of Himachal Pradesh Vs. Jai Lal and Others[16]: In this case, the court underscored the importance of cross-examining expert witnesses, alongside their responsibility to provide expert opinions.

Prahlad Saran Gupta v Bar Council of India[17]: This case emphasised that there is no need for additional confirmation once an opinion from a handwriting expert is provided. This case discussed the weight and importance attributed to expert opinions, particularly in matters related to signatures and handwriting.

B. *Other Important Observations*

Gautam Kundu v. State of West Bengal[18]*:* In this case, it was established that Indian courts cannot order blood tests routinely or without proper justification. Such requests should not be allowed as a means of conducting a general or unfocused inquiry. There must be a strong prima facie case, and the court must carefully consider the potential consequences before directing a blood test.

Sharda v. Dharmpal[19]*:* The court held that a matrimonial court has the authority to order medical tests, including DNA tests, but this power should be exercised only when the applicant presents robust prima facie case which must be reinforced by sufficient evidence.

Conclusion

Experts' opinion plays a vital role in the Indian judicial scheme. They help understand and clarify complex issues which requires special knowledge.

The are rules of prudence ensuring that an expert's opinion is reliable, such as an expert's testimony in divorce or bigamy cases having strong evidentiary support and the expert having other pieces of evidence to back their claim.

The decision in *S. Gopala Reddy vs State of A.P.*[20] demonstrates these precautionary measures. "While Expert's Opinions are indeed helpful, they

[14] (1994) 4 SCC 223
[15] (1992) 2 SCC 50
[16] (1999) 7 SCC 280
[17] AIR 1997 SC 1338
[18] 1993 SCR (3) 917
[19] AIR 2003 SC 3450
[20] (1996) 4 SCC 596

should be accompanied by other forms of evidence in order for the legal system to remain untainted and reliable."

As in any case, the burden of proof (Section 101), and the phrasing "beyond reasonable doubt" followed in criminal cases, and "preponderance of probabilities" for civil cases determines the ultimate judgement.

Case Laws

1. *Daubert v. Merrell Dow Pharmaceuticals, Inc.*, 509 U.S. 579 (1993).
2. *State of Himachal Pradesh vs. Jai Lal and Others*, Supreme Court of India.
3. *Ramesh Chandra Agrawal v. Regency Hospital Ltd & Ors.*, Supreme Court of India.
4. *Malay Kumar Ganguly v. Dr. Sukumar Mukherjee*, Supreme Court of India.
5. *Bant Singh v. Niranjan Singh*, Supreme Court of India.
6. *State of Kerala v. Mahesh*, Supreme Court of India.
7. *Gade Lakshmi Mangaraju v. State of Andhra Pradesh*, Supreme Court of India.
8. *Prahlad Saran Gupta v. Bar Council of India*, Supreme Court of India.
9. *Gautam Kundu v. State of West Bengal*, Supreme Court of India.
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2. The Code of Criminal Procedure, 1973.
3. Information Technology Act, 2000.
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